Name:harish Panpaliya

Branh:CS_AI

Division: D Batch-3

Roll no:52

PRN no: 12412173

Date: 08/01/2026

Time: 4:20

Kaggle dataset link: https://www.kaggle.com/datasets/lashagoch/life-expectancy-who-updated

Perform the following operations using Python on suitable data sets:

1.Read data from different formats (like csv, xls)

2.Find Shape of Data

3.Find Missing Values

4.Find data type of each column

5.Finding out Zero's

6.Indexing and selecting data, sort data,

7.Describe attributes of data, checking data types of each column,

8.counting unique values of data, format of each column, converting variable data type (e.g. from long to short, vice versa)

Imported Pandas and numpy

```
import pandas as pd
import numpy as np
```

Read The CSv and Xlsx file and print them

```
df=pd.read_csv('dataset.csv')
```

```
print(df)
```

```
1              57.9025                10.350            97        94   26.0
```

```
2           201.0765                1.570           60       35  21.2  ...
3           222.1965                5.680           93       74  25.3  ...
4            57.9510                2.890           97       89  27.0  ...
...              ...                  ...          ...      ...   ...  ...
2859         291.8240                0.092           72       64  20.8  ...
2860         235.2330                6.560           97       97  25.3  ...
2861         134.8950                1.560           62       95  21.9  ...
2862         204.0120               11.000           94       95  26.1  ...
2863          50.5745                6.840           88       90  26.1  ...

      Diphtheria  Incidents_HIV  GDP_per_capita  Population_mln  \
0             97           0.08           11006           78.53
1             97           0.09           25742           46.44
2             64           0.13            1076         1183.21
3             93           0.79            4146            0.75
4             94           0.08           33995            7.91
...          ...            ...             ...             ...
2859          34           0.49             399           11.33
2860          95           0.02            2515            2.67
2861          97           0.02            2130           19.39
2862          95           0.05            7424            3.44
2863          95           0.05           48934            0.32

      Thinness_ten_nineteen_years  Thinness_five_nine_years  Schooling  \
0                             4.9                       4.8        7.8
1                             0.6                       0.5        9.7
2                            27.1                      28.0        5.0
3                             5.7                       5.5        7.9
4                             1.2                       1.1       12.8
...                           ...                       ...        ...
2859                         12.8                      12.9        1.1
2860                          2.2                       2.3        9.1
2861                         15.4                      15.5       10.3
2862                          3.3                       3.3       11.1
2863                          0.9                       0.9       11.0

      Economy_status_Developed  Economy_status_Developing  Life_expectancy
0                            0                          1             76.5
1                            1                          0             82.8
2                            0                          1             65.4
3                            0                          1             67.0
4                            1                          0             81.7
...                        ...                        ...              ...
2859                         0                          1             49.9
2860                         0                          1             66.9
2861                         0                          1             74.3
2862                         1                          0             71.8
2863                         1                          0             82.4

[2864 rows x 21 columns]
```

```
df
```

	Country	Region	Year	Infant_deaths	Under_five_deaths	Adult_mortality	Alcohol_consumption	Hepatitis_B	Measles
0	Turkiye	Middle East	2015	11.1	13.0	105.8240	1.320	97	65
1	Spain	European Union	2015	2.7	3.3	57.9025	10.350	97	94
2	India	Asia	2007	51.5	67.9	201.0765	1.570	60	35
3	Guyana	South America	2006	32.8	40.5	222.1965	5.680	93	74
4	Israel	Middle East	2012	3.4	4.3	57.9510	2.890	97	89
...	...	...	...	...	...	...	...	...	...
2859	Niger	Africa	2000	97.0	224.9	291.8240	0.092	72	64
2860	Mongolia	Asia	2009	23.9	28.6	235.2330	6.560	97	97
2861	Sri Lanka	Asia	2004	17.7	28.9	134.8950	1.560	62	95
2862	Lithuania	European Union	2002	7.9	9.9	204.0120	11.000	94	95
2863	Iceland	Rest of Europe	2011	2.1	2.6	50.5745	6.840	88	90

2864 rows × 21 columns

```
df_xl=pd.read_excel('dataset2.xlsx')
```

```
print(df_xl)
```

```
2860   Mongolia            Asia  2009           23.9               28.6
2861  Sri Lanka            Asia  2004           17.7               28.9
2862  Lithuania  European Union  2002            7.9                9.9
2863    Iceland  Rest of Europe  2011            2.1                2.6

      Adult_mortality  Alcohol_consumption  Hepatitis_B  Measles   BMI  ...  \
0            105.8240                1.320           97       65  27.8  ...
1             57.9025               10.350           97       94  26.0  ...
2            201.0765                1.570           60       35  21.2  ...
3            222.1965                5.680           93       74  25.3  ...
4             57.9510                2.890           97       89  27.0  ...
...               ...                  ...          ...      ...   ...  ...
2859         291.8240                0.092           72       64  20.8  ...
2860         235.2330                6.560           97       97  25.3  ...
2861         134.8950                1.560           62       95  21.9  ...
2862         204.0120               11.000           94       95  26.1  ...
2863          50.5745                6.840           88       90  26.1  ...

      Diphtheria  Incidents_HIV  GDP_per_capita  Population_mln  \
0             97           0.08           11006           78.53
1             97           0.09           25742           46.44
2             64           0.13            1076         1183.21
3             93           0.79            4146            0.75
4             94           0.08           33995            7.91
...          ...            ...             ...             ...
2859          34           0.49             399           11.33
2860          95           0.02            2515            2.67
2861          97           0.02            2130           19.39
2862          95           0.05            7424            3.44
2863          95           0.05           48934            0.32

      Thinness_ten_nineteen_years  Thinness_five_nine_years  Schooling  \
0                             4.9                       4.8        7.8
1                             0.6                       0.5        9.7
2                            27.1                      28.0        5.0
3                             5.7                       5.5        7.9
4                             1.2                       1.1       12.8
...                           ...                       ...        ...
2859                         12.8                      12.9        1.1
2860                          2.2                       2.3        9.1
2861                         15.4                      15.5       10.3
2862                          3.3                       3.3       11.1
2863                          0.9                       0.9       11.0

      Economy_status_Developed  Economy_status_Developing  Life_expectancy
0                            0                          1             76.5
1                            1                          0             82.8
2                            0                          1             65.4
3                            0                          1             67.0
4                            1                          0             81.7
...                        ...                        ...              ...
2859                         0                          1             49.9
2860                         0                          1             66.9
2861                         0                          1             74.3
2862                         1                          0             71.8
2863                         1                          0             82.4

[2864 rows x 21 columns]
```

```
df_xl
```

	Country	Region	Year	Infant_deaths	Under_five_deaths	Adult_mortality	Alcohol_consumption	Hepatitis_B	Measles
0	Turkiye	Middle East	2015	11.1	13.0	105.8240	1.320	97	65
1	Spain	European Union	2015	2.7	3.3	57.9025	10.350	97	94
2	India	Asia	2007	51.5	67.9	201.0765	1.570	60	35
3	Guyana	South America	2006	32.8	40.5	222.1965	5.680	93	74
4	Israel	East	2012	3.4	4.3	57.9510	2.890	97	89
				...	...	...	...	...	...
2859	Niger	Africa	2000	97.0	224.9	291.8240	0.092	72	64
2860	Mongolia	Asia	2000	23.9	28.6	235.2330	6.560	97	97
2861	Sri Lanka	Asia	2004	17.7	28.9	134.8950	1.560	62	95
2862	Zimbabwe	Union	[illegible]	7.9	9.9	204.0120	11.000	94	95
				2.1	2.6	50.5745	6.840	88	90

2864 rows × 21 columns

datatype of data in each column

```
df.dtypes
```

	0
Country	object
Region	object
Year	int64
Infant_deaths	float64
Under_five_deaths	float64
Adult_mortality	float64
Alcohol_consumption	float64
Hepatitis_B	int64
Measles	int64
BMI	float64
Polio	int64
Diphtheria	int64
Incidents_HIV	float64
GDP_per_capita	int64
Population_mln	float64
Thinness_ten_nineteen_years	float64
Thinness_five_nine_years	float64
Schooling	float64
Economy_status_Developed	int64
Economy_status_Developing	int64
Life_expectancy	float64

dtype: object

shape of dataset

```
df.shape
```

```
(2864, 21)
```

Reading head and Tail of the data

```
df.head()
```

	Country	Region	Year	Infant_deaths	Under_five_deaths	Adult_mortality	Alcohol_consumption	Hepatitis_B	Measles	BM
0	Turkiye	Middle	2015	11.1	13.0	105.8240	1.32	97	65	27.
1	Spain	European Union	2015	2.7	3.3	57.9025	10.35	97	94	26.
3	Guyana	America	2006	32.8	40.5	222.1965	5.68	93	74	25.
4	Israel	Middle	2013	3.4	4.3	57.0510	2.80	97	80	27.

```
df.tail(3)
```

	Country	Region	Year	Infant_deaths	Under_five_deaths	Adult_mortality	Alcohol_consumption	Hepatitis_B	Measles
2861	Sri Lanka	Asia	2004	17.7	28.9	134.8950	1.56	62	95
2862	Lithuania	Rest of Europe	2002	7.9	9.9	204.0120	11.00	94	95
2863	Iceland	Rest of Europe	2011	2.1	2.6	50.5745	6.84	88	90

3 rows × 21 columns

```
df.index
```

```
RangeIndex(start=0, stop=2864, step=1)
```

```
df.describe()
```

	Year	Infant_deaths	Under_five_deaths	Adult_mortality	Alcohol_consumption	Hepatitis_B	Measles	
count	2864.000000	2864.000000	2864.000000	2864.000000	2864.000000	2864.000000	2864.000000	2864.00
mean	2007.500000	30.363792	42.938268	192.251775	4.820882	84.292598	77.344972	25.03
std	4.610577	27.538117	44.569974	114.910281	3.981949	15.995511	18.659693	2.19
min	2000.000000	1.800000	2.300000	49.384000	0.000000	12.000000	10.000000	19.80
25%	2003.750000	8.100000	9.675000	106.910250	1.200000	78.000000	64.000000	23.20
50%	2007.500000	19.600000	23.100000	163.841500	4.020000	89.000000	83.000000	25.50
75%	2011.250000	47.350000	66.000000	246.791375	7.777500	96.000000	93.000000	26.40
max	2015.000000	138.100000	224.900000	719.360500	17.870000	99.000000	99.000000	32.10

sort by year

```
df.sort_values(by="Year")
```

	Country	Region	Year	Infant_deaths	Under_five_deaths	Adult_mortality	Alcohol_consumption	Hepatitis_B	Measle
744	Eritrea	Africa	2000	56.1	85.2	370.4550	0.88	87	8
7	Hungary	European Union	2000	8.7	10.1	192.9690	12.23	88	9
1789	North Macedonia	Rest of Europe	2000	14.3	16.0	123.0060	4.49	72	9
2402	Belgium	European Union	2000	4.8	5.8	99.6000	11.21	60	7
2823	Czechia	European Union	2000	4.5	5.5	124.3800	13.90	92	9
...	...	...	...	...	...	...	...	...	.
1	Spain	European Union	2015	2.7	3.3	57.9025	10.35	97	9
6	Russian Federation	Rest of Europe	2015	6.6	8.2	223.0000	8.06	97	9
2848	Bahrain	Middle East	2015	6.5	7.6	66.6840	1.53	98	9
27	Cameroon	Africa	2015	57.0	88.0	340.1265	4.55	84	6
2820	Venezuela, RB	South America	2015	16.4	19.1	141.5540	6.21	87	8

2864 rows × 21 columns

sort by axis

```
df.sort_index(axis=1)
```

	Adult_mortality	Alcohol_consumption	BMI	Country	Diphtheria	Economy_status_Developed	Economy_status_Developing
0	105.8240	1.320	27.8	Turkiye	97	0	1
1	57.9025	10.350	26.0	Spain	97	1	0
2	201.0765	1.570	21.2	India	64	0	1
3	222.1965	5.680	25.3	Guyana	93	0	1
4	57.9510	2.890	27.0	Israel	94	1	0
...	...	...	...	...	...	...	...
2859	291.8240	0.092	20.8	Niger	34	0	1
2860	235.2330	6.560	25.3	Mongolia	95	0	1
2861	134.8950	1.560	21.9	Sri Lanka	97	0	1
2862	204.0120	11.000	26.1	Lithuania	95	1	0
2863	50.5745	6.840	26.1	Iceland	95	1	0

2864 rows × 21 columns

Created own data and dataframe

```
from numpy import nan
import pandas as pd
data={
    'Name':["Harish","Ram","shital","Riya","OM","Rohan"],
    'Age':[18,19,20,21,18,18],
    'Gender':['M','M','F','F','M','M'],
    'Marks':[90,80,50,40,nan,75],
    'Internship':[1.0,0.0,1.0,1.0,0.0,0.0]
}
df_f=pd.DataFrame(data)
df_f
```

	Name	Age	Gender	Marks	Internship
0	Harish	18	M	90.0	1.0
1	Ram	19	M	80.0	0.0
2	shital	20	F	50.0	1.0
3	Riya	21	F	40.0	1.0
4	OM	18	M	NaN	0.0
5	Rohan	18	M	75.0	0.0

Next steps: Generate code with df_f New interactive sheet

datatypes of data in dataframe

```
from numpy import nan
import pandas as pd
data={
    'Name':["Harish","Ram","shital","Riya","OM","Rohan"],
    'Age':[18,19,20,21,18,18],
    'Gender':['M','M','F','F','M','M'],
    'Marks':[90,80,50,40,nan,75],
    'Internship':[1.0,0.0,1.0,1.0,0.0,0.0]
}
df_f=pd.DataFrame(data)
```

```
df_f.dtypes
```

	0
Name	object
Age	int64
Gender	object
Marks	float64
Internship	float64

dtype: object

Index of row with marks=NULL

```
from numpy import nan
import pandas as pd
data={
    'Name':["Harish","Ram","shital","Riya","OM","Rohan"],
    'Age':[18,19,20,21,18,18],
    'Gender':['M','M','F','F','M','M'],
    'Marks':[90,80,50,40,nan,75],
    'Internship':[1.0,0.0,1.0,1.0,0.0,0.0]
}
df_f=pd.DataFrame(data)
nan_indices = df_f[df_f['Marks'].isna()].index.tolist()
nan_indices
```

```
[4]
```

Drop rows which have null value --> temporary

```
from numpy import nan
import pandas as pd
data={
    'Name':["Harish","Ram","shital","Riya","OM","Rohan"],
    'Age':[18,19,20,21,18,18],
    'Gender':['M','M','F','F','M','M'],
    'Marks':[90,80,50,40,nan,75],
    'Internship':[1.0,0.0,1.0,1.0,0.0,0.0]
}
df_f=pd.DataFrame(data)
df_f.dropna(how="any")
```

	Name	Age	Gender	Marks	Internship
0	Harish	18	M	90.0	1.0
1	Ram	19	M	80.0	0.0
2	shital	20	F	50.0	1.0
3	Riya	21	F	40.0	1.0
5	Rohan	18	M	75.0	0.0

using ioc,iloc,at and isin

```
# df_f.loc("Age",)=[20,22,25,18,18,30]
# The correct syntax
df_f.loc[ :,"Age"] = [20, 22, 25, 18,30,18]
df_f
```

	Name	Age	Gender	Marks	Internship
0	Harish	20	M	90.0	1.0
1	Ram	22	M	80.0	0.0
2	shital	25	F	50.0	1.0
4	OM	20	M	NaN	0.0
5	Rohan	18	M	75.0	0.0

```
df_f.iloc[[0, 2, 5], [1, 3]]
```

	Age	Marks
0	20	90.0
2	25	50.0
5	18	75.0

Next steps: Generate code with df_f New interactive sheet

```
mark = df_f.at[5, 'Marks']
print(mark)
```

```
75.0
```

```
df_f[df_f["Name"].isin(["Harish","Ram","shital","Riya","OM","Rohan"])]
```

	Name	Age	Gender	Marks	Internship
0	Harish	18	M	90.0	1.0
1	Ram	19	M	80.0	0.0
2	shital	20	F	50.0	1.0
3	Riya	21	F	40.0	1.0
4	OM	18	M	NaN	0.0
5	Rohan	18	M	75.0	0.0

```
df_f["Name"].isin(["Harish","Ram","shital","Riya","OM","Rohan"])
```

	Name
0	True
1	True
2	True
3	True
4	True
5	True

dtype: bool

calculated Average of marks excluding null

```
from numpy import nan
import pandas as pd
data={
    'Name':["Harish","Ram","shital","Riya","OM","Rohan"],
    'Age':[18,19,20,21,18,18],
    'Gender':['M','M','F','F','M','M'],
    'Marks':[90,80,50,40,nan,75],
    'Internship':[1.0,0.0,1.0,1.0,0.0,0.0]
}
df_f=pd.DataFrame(data)
df_f.dropna(how="any")['Marks'].mean()
```

```
np.float64(67.0)
```

```
average_marks = df_f['Marks'].mean()
average_marks
```

```
np.float64(67.0)
```

```
c=avg=0
for ele in df_f['Marks']:
```

```
    if pd.notnull(ele):
        c=c+1
        avg=avg+ele
avg=avg/c
print(avg)
```

```
67.0
```

Changed Male to 0 and Female to 1

```
from numpy import nan
import pandas as pd
data={
    'Name':["Harish","Ram","shital","Riya","OM","Rohan"],
    'Age':[18,19,20,21,18,18],
    'Gender':['M','M','F','F','M','M'],
    'Marks':[90,80,50,40,nan,75],
    'Internship':[1.0,0.0,1.0,1.0,0.0,0.0]
}
df_f=pd.DataFrame(data)
df_f['Gender'].map({'M':0,'F':1,}).astype(float)
```

	Gender
0	0.0
1	0.0
2	1.0
3	1.0
4	0.0
5	0.0

dtype: float64

user define function using agg

```
# This returns the new Age column
df_f['Age'].agg(lambda x: x + 1)
```

```
/tmp/ipython-input-3236170448.py:2: FutureWarning: using <function <lambda> at 0x7ddaa66e96c0> in Series.agg cannot aggregat
  df_f['Age'].agg(lambda x: x + 1)
```

	Age
0	19
1	20
2	21
3	22
4	19
5	19

dtype: int64

Sorted students who scored more than or equal to 80

```
df_s=df_f[df_f['Marks']>=80].copy()
df_s
```

	Name	Age	Gender	Marks	Internship
0	Harish	18	M	90.0	1.0
1	Ram	19	M	80.0	0.0

Next steps: Generate code with df_s New interactive sheet

Removed Age column

```
df_f.drop('Age',axis=1, inplace=True)
df_f
```

	Name	Gender	Marks	Internship
0	Harish	M	90.0	1.0
1	Ram	M	80.0	0.0
2	shital	F	50.0	1.0
3	Riya	F	40.0	1.0
4	OM	M	NaN	0.0
5	Rohan	M	75.0	0.0

Next steps: Generate code with df_f New interactive sheet

Slicing

```
import pandas as pd
from numpy import nan
data={
    'Name':["Harish","Ram","shital","Riya","OM","Rohan"],
    'Age':[18,19,20,21,18,18],
    'Gender':['M','M','F','F','M','M'],
    'Marks':[90,80,50,40,nan,75],
    'Internship':[1.0,0.0,1.0,1.0,0.0,0.0]
}
df_f=pd.DataFrame(data)
df_f.iloc[0:4]
```

	Name	Age	Gender	Marks	Internship
0	Harish	18	M	90.0	1.0
1	Ram	19	M	80.0	0.0
2	shital	20	F	50.0	1.0
3	Riya	21	F	40.0	1.0

```
df_f.sort_values(by='Name')
```

	Name	Age	Gender	Marks	Internship
0	Harish	18	M	90.0	1.0
4	OM	18	M	NaN	0.0
1	Ram	19	M	80.0	0.0
3	Riya	21	F	40.0	1.0
5	Rohan	18	M	75.0	0.0
2	shital	20	F	50.0	1.0

```
df_f.fillna(60.0)
```

	Name	Age	Gender	Marks	Internship
0	Harish	18	M	90.0	1.0
1	Ram	19	M	80.0	0.0
2	shital	20	F	50.0	1.0
3	Riya	21	F	40.0	1.0
4	OM	18	M	60.0	0.0
5	Rohan	18	M	75.0	0.0

```
data2={
    'Name':["Harish","Ram","shital","Riya","OM","Rohan"],
    'Pending_fee':[1,0,0,1,1,0],
```

```
    'Branch':['CSE','CSE_AI','AI_Ml','CSE','CSE','CSE'],
    'rating':[5,3,2,4,1,5]


}
```

```
df_f=pd.DataFrame(data2)
```

```
df_f["Branch"]
```

	Branch
0	CSE
1	CSE_AI
2	AI_Ml
3	CSE
4	CSE
5	CSE

dtype: object

```
df_f.loc[:,["Name","Pending_fee"]]
```

	Name	Pending_fee
0	Harish	1
1	Ram	0
2	shital	0
3	Riya	1
4	OM	1
5	Rohan	0

```
df_f.iloc[3]
```

	3
Name	Riya
Pending_fee	1
Branch	CSE
rating	4

dtype: object

```
s1 = pd.Series(
   [1, 2, 3, 4, 5, 6],
   index=pd.date_range("20260122", periods=6))
s1
```

	0
2026-01-22	1
2026-01-23	2
2026-01-24	3
2026-01-25	4
2026-01-26	5
2026-01-27	6

dtype: int64